Region Based Resolution Enhanced Sequential to Segregated Dual Tiny Multi Scale Attention Net Based OCTA Image Segmentation

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***Abstract*—OCTA is an important imaging modality used to view small retinal blood vessels in a high-resolution, dye-free, and non-invasive manner. As its clinical use increases, the need for accurate and reliable automated vessel segmentation will be crucial for proper diagnosis and measurement. However, even with the rich vascular details present in OCTA images, they remain challenging to segment due to several limiting factors such as noise [1], [2], low contrast, and complex structure of fine capillaries. This study introduces TinyMSANet, a Region-Based Resolution-Enhanced Sequential to Segregated Dual Tiny Multi-Scale Attention Network for efficient and precise OCTA image segmentation. The framework first conducts an Enhanced Super-Resolution GAN (ESRGAN) to improve low-resolution OCTA patches by recovering vessel details and clarity. Then a dual-stage TinyMSANet performs vessel segmentation and a refinement step for complex artery–vein pattern discrimination. Tests on the OCTA-500 dataset showed that compared with the CNN-based method [3], U-Net, and other attention models, the better performance achieved a Dice of 0.76 and PSNR of 20.27dB. With this lightweight architecture and strong results, the system scales up to a real-time solution for advanced retinal vessel analysis for clinical necessity.**

Keywords: Optical Coherence Tomography Angiography(OCTA), Multi-scale attention(MSA), ESRGAN(Enhanced Super-Resolution Generative Adversarial Network

I. Introduction

OCTA has emerged as an important imaging tool in modern eye care [4]–[6]. It offers a non-invasive method for visualization of small retinal blood vessels at higher resolution. Without injecting any dye, OCTA provides depth-dependent flow information that assists clinicians in detecting early signs of various diseases such as diabetic retinopathy, glaucoma, and age-related macular degeneration. Its rapidly increasing clinical applications create an urgent need for computer-aided diagnosis tools to extract vascular features from OCTA images with great accuracy. With its advantages, OCTA still faces several challenges that make the accurate segmentation of vessels difficult. Traditional methods based on mathematical rules generally provide poor outcomes due to the difficulties in handling projection artifacts, motion noise, uneven lighting, and complex capillary patterns. In general, deep learning approaches are more powerful but usually computationally expensive and slow; thus, they are not suitable for real-time application in clinics. Also, both traditional and deep learning approaches struggle because retinal vessels are tightly packed, with very low contrast between arteries and veins. As a result, they often fail to keep fine branches, separate overlapping vessels, or work well across datasets from different scanners. These issues limit the use of automated OCTA analysis in everyday eye-care practice.

To solve these challenges, this paper presents a region-focused, resolution-enhanced framework for accurate retinal vessel reconstruction and segmentation using lightweight deep learning modules. The method uses an Enhanced Super-Resolution GAN to selectively improve important retinal regions by making tiny blood vessels clearer. The improved images are then passed to TinyMSANet, a Dual Tiny Multi-Scale Attention Network, which is a lightweight model designed to detect fine capillaries while still running fast enough for real-time use [7].

- ESRGAN for Resolution Enhancement: It helps to overcome the low-resolution OCTA inputs by restoring fine capillary structures and improving vessel clarity of any smaller region of $64 \times 64 \times 3$ image.
- TinyMSANet for Lightweight Segmentation: Designed to provide efficient vessel segmentation with minimal parameters, enabling faster inference for clinical use.
- Stacked TinyMSANet for Structural Refinement: It improves the segmentation accuracy by refining multi-scale vascular features and recovering thin, intertwined vessels.
- Artery–Vein Differentiation and Noise Robustness: Fixing low contrast, overlapping vessels, and noise artifacts to enable better separation of arteries and veins.

II. Related Work

This section briefly discusses some research on the segmentation of retinal blood vessels in OCTA images. Hao et al. [8] developed a sparse-based domain adaptation network called SASR that enhances low-resolution OCTA images to reduce the difference between synthetic and real data through adversarial learning and sparse edge-aware loss. Their proposed method has improved vessel edges while boosting accuracy in the segmentation task. The quality of images has also

been enhanced based on PSNR, SSIM, and LPIPS scores, which in turn makes clinical measurement of vascular structure much easier and supports better diagnosis of retinal diseases. Manassakorn et al. [9] proposed GlauNet, a CNN model for detecting glaucoma automatically from OCTA images. The proposed network consists of three convolutional layers and five fully connected layers. It achieved a high sensitivity of 88.9% and specificity of 89.6% on mixed glaucoma datasets and outperformed models such as VGG16 and ResNet50. It is also robust to various imaging artifacts and has the potential to enhance clinical diagnosis.

Li et al. [10] proposed the Image Projection Network, an end-to-end approach to transform 3D OCTA images into 2D maps with one-direction pooling inside a Projection Learning Module. IPN achieved high accuracy and stability without requiring any retinal layer segmentation, thus offering better diagnosis of eye diseases [11]. Zhang et al. [12] developed a fast artifact-suppression OCTA approach by multiplying the correlation OCTA signals with squared structural images, and then performing second-order temporal differences to suppress the noise. This method eliminates shadows and artifacts while preserving vessel information, with computation about tenfold simpler than the complex approaches, enabling near real-time imaging. Huang et al. [13] proposed PMFAN, a patch-based, unsupervised 3D deep learning architecture for generating cross-device OCTA by mapping OCT data to a shared feature space with preserved fine vessel details and mitigated boundary artifacts. Compared with the state-of-the-art methods, their approach achieved better vascular visualization and metric accuracy, enabling the analysis of retinal diseases in cases where OCTA hardware was not available.

Mao et al. [14] proposed MR2-Net to address limited OCTA field-of-view by fusing neighboring scans while retaining resolution and vascular integrity. The two-stage registration and fusion utilized Progressive Multi-Resolution Feature Fusion, Region-Guided Dynamic Localization, and Vessel Alignment Loss, delivering superior SSIM and clinically validated image quality compared to classical and deep learning methods.

Hao et al. [15] developed VAFF-Net, a multi-task deep learning network for simultaneous detection, segmentation, and classification of retinal vessels, foveal avascular zones, and vessel junctions in OCTA images. Utilizing depth-resolved en face angiograms and adaptive Voting Gate Modules, VAFF-Net outperformed state-of-the-art approaches with superior vessel SSIM and junction classification, generalizing well to color fundus images.

Jiang et al. [16] introduced PL-Joint-Seg, a two-stream network jointly segmenting foveal avascular zones and retinal vessels leveraging 3D projection learning and 2D spatial features fused via bottleneck attention modules. The method achieved state-of-the-art Dice scores on OCTA-500 datasets, with enhanced boundary delineation and noise robustness.

Raja et al. [17] presented Dense-PMSFNet, a DenseNet-based deep learning model for accurate segmentation of retinal capillaries, arteries, veins, and foveal avascular zones in OCTA images. Utilizing multi-scale pyramidal fusion and squeeze-and-excitation blocks, their model achieved superior Dice and IoU metrics, proving effective for diabetic retinopathy diagnosis.

Abdelsalam et al. [18] proposed a multifractal geometry analysis with Support Vector Machine classification for early diabetic retinopathy detection from OCTA macular images. Leveraging multifractal features extracted via fractal analysis, their method achieved 98.5% classification accuracy, and clinical promise for non-invasive early screening.

The complete related work survey has been summarized in Table I.

Most of the current OCTA segmentation methods face challenges in terms of small datasets, high computational requirements, and generalizing to noisy and fine small vessel details. Moreover, complex vessel structures remain challenging to align in stitched images. More often than not, these conditions reduce accuracy and limit use in real clinical settings. This indicates the need for a method that would better focus attention on important areas, improve image quality where necessary, and capture fine vascular features more effectively while keeping computation manageable. The main contributions of this work include:

- Artery–Vein Discrimination: Attention is given to the accurate segmentation of artery and vein structures in OCTA images, enabling a more detailed vessel-level analysis than most current OCTA methods can provide.
- High-Resolution Regional Enhancement: It applies a super-resolution module selectively, enhancing the important retinal areas so that vessels become clearer while keeping the original structure intact.
- Lightweight Architecture for Real-Time Use: A small and efficient TinyMSANet model, comprising a dual-stage design for real-time clinical use that provides strong performance with much lower computation

III. Proposed Methodology

The main objective of this project is to increase the speed and accuracy of OCTA vessel map segmentation for whatever region of the image is necessary. Instead, the proposed method looks on the key vascular areas containing arteries and veins. Selective enhancement of the clarity and contrast of focused regions before segmentation allows for better medium-sized vessel detection by the system. This region-focused approach optimizes computational efficiency while yielding more reliable and precise vessel segmentation results, eventually enhancing vascular analysis in real time. The following subsection provides details about the dataset used to develop and evaluate the proposed approach shown in fig 1.

A. Dataset

This study is performed over 2D en face OCTA images from the publicly available OCTA-500 dataset. Optical Coherence Tomography Angiography (OCTA) enables the visualization of retinal blood flow without the need for contrast dyes. It is formed by repeatedly scanning the same retinal location to detect motion signals generated by moving red blood cells. The technique operates on both spectral-domain and swept-source

TABLE I: Related Work Survey

Author	Year	Approach	Evaluation Matrices	Results	Observation	Limitation
Hao et al. [8]	2022	Sparse-based domain adaptation with GAN and edge loss	PSNR, SSIM, LPIPS, Dice, AUC, SEN	Dice 0.601, Auc 0.747	Good vessel edge details, helps segmentation	Struggles with perfect registration, complex vessels
Manassakorn et al. [9]	2022	Custom CNN, GlauNet for glaucoma detection in OCTA	Sensitivity, Specificity, AUC, Accuracy	Sensitivity 88.9%, Specificity 89.6%, AUC 0.89	Robust to artifacts, accurate in small datasets	Dataset is small, trained on single device
Li et al. [10]	2025	3D-to-2D deep learning with projection learning	Dice, Jaccard, Precision, Recall	Dice 0.88 for vessels, 0.89 for FAZ	No need for retinal layer segmentation	Needs a lot of computing, FAZ location limits
Huang et al. [13]	2024	Unsupervised patch-based 3D domain adaptation	SVA, CNR, Accuracy, Dice, ROC	Best cross-device OCTA generation performance	Generalizes to new devices, helps disease diagnosis	Misses fine capillaries, resolution issues
Zhang et al. [12]	2025	Combines OCT	OCTA for fast artifact removal	SNR, artifact ratio, speed, vessel overlap	PSNR 16db, artifacr ratio 76%	Less accurate, sensitive to noise
Hao et al. [15]	2022	VAFF-Net with multi-input en face slabs	Dice, IoU, Sensitivity	Dice 0.81	Multi-input slabs improve accuracy	Cross-device domain shift
Mao et al. [14]	2025	MR2-Net two-step image stitching	SSIM, Similarity metrics	Outperformed existing methods	Superior stitching performance	Domain shift robustness issues
Jiang et al. [16]	2025	PL-Joint-Seg dual-path framework	Dice coefficient	State-of-the-art performance	Joint FAZ and RV segmentation	High computational complexity
Raja et al. [17]	2025	Dense-PMSFNet (DenseNet-121)	Dice, Sensitivity, Specificity	Superior segmentation performance	Multi-scale feature extraction	High computational cost
Abdelsalam et al. [18]	2021	Multifractal geometry + SVM	Accuracy, Sensitivity, Specificity	98.5% accuracy achieved	Effective early DR detection	Image quality dependency

OCT systems, providing a completely non-invasive method for high-resolution vascular imaging. By applying motion contrast techniques such as decorrelation or phase-variance, OCTA can rapidly capture and reconstruct detailed vascular networks across multiple retinal layers. In most commercial OCTA systems, multiple B-scans are acquired per location to quantify blood flow, which are then processed to generate clear 2D en face projection images.

Each image in the OCTA-500 dataset has a spatial resolution of $400 \times 400 \times 3$ shown in fig 2, providing sufficient detail for both large and fine vessel structures. The next subsection tells about the preprocessing techniques applied to these images to enhance vessel visibility and prepare them for efficient segmentation.

B. Data Preprocessing

In the preprocessing stage, regions surrounding the foveal avascular zone and other densely vascularized areas were cropped to emphasize clinically significant structures while eliminating background noise and irrelevant regions. Each cropped image are of resolution $256 \times 256 \times 3$ shown in fig 3.

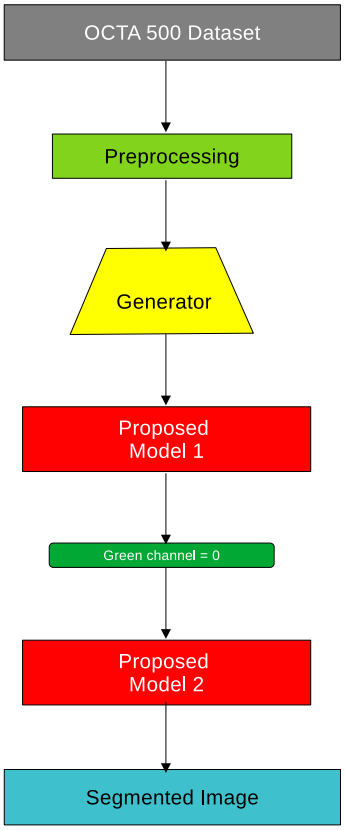


Fig. 1: Approach.

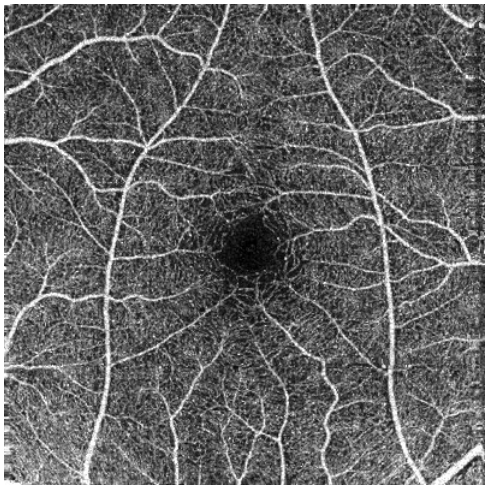


Fig. 2: $400 \times 400 \times 3$ OCTA image sample.

To further enhance vessel visibility, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied with a tile size of 8, improving local contrast and revealing fine vascular structures without amplifying noise. Finally, the enhanced images were downsampled to $64 \times 64 \times 3$ shown in fig 4.

These preprocessing steps effectively emphasized the vascular details, enabling the segmentation model to learn precise and discriminative vessel patterns. The architectural flow diagram is shown in fig 6b The following section describes the proposed approach used for vessel segmentation.

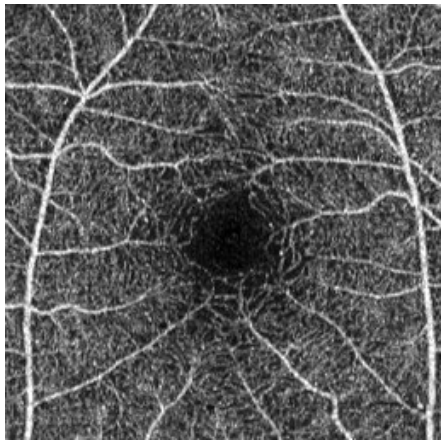


Fig. 3: $256 \times 256 \times 3$ OCTA image sample.

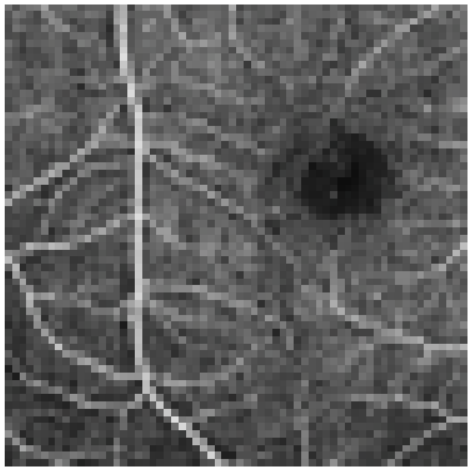


Fig. 4: $64 \times 64 \times 3$ OCTA image sample.

C. Enhanced Super-Resolution Generative Adversarial Network

For resolution enhancement, after performing on several GAN [19]–[21]-based super-resolution methods to improve the visual quality of medical images and selected the Enhanced Super-Resolution Generative Adversarial Network (ESRGAN) [22] for this task. ESRGAN is used to upsample the cropped $64 \times 64 \times 3$ CLAHE-enhanced OCTA patches into its specific high-resolution $256 \times 256 \times 3$ images, makes finer visualization of small and medium-sized retinal vessels. Unlike conventional interpolation techniques that often blur intricate details, ESRGAN utilize both adversarial loss, perceptual content loss and pixel loss, allowing it to reconstruct sharper textures and retain subtle vascular structures. This enhancement step will ensure that even thin and complex vessel patterns are more distinguishable, providing a clearer and more informative input for the subsequent segmentation process. The following

generator and discriminator is shown in fig 5a and 5b. And the fig 7 shows the adversarial way of training ESRGAN.

The following section presents the proposed model for vessel segmentation, which leverages these enhanced images to achieve precise and efficient delineation of retinal vasculation.

Algorithm 1 Relativistic Patch-based Discriminator

procedure ESRGANDISCRIMINATOR($x, ; in_n c = 3, ; base_n f = 64$)
  $in_f \leftarrow in_n c$
  $nf \leftarrow base_n f$
  for $i = 1$ **to** 6 **do**
    if $i < 6$ **then**
      $out_f \leftarrow nf$
    else
      $out_f \leftarrow 2 \times nf$
    end if
    $stride \leftarrow 1$ **if** (i is odd) **else** 2
    $x \leftarrow$ Conv2D($x, ; in_c hannels = in_f, ; out_c hannels = out_f, ; kernel = 3 \times 3, ; stride = stride, ; padding = 1$)
    $x \leftarrow$ BatchNorm(x)
    $x \leftarrow$ LeakyReLU($x, ; slope = 0.2$)
    $in_f \leftarrow out_f$
    if i is even **then**
      $nf \leftarrow 2 \times nf$ ▷ double base filters every two blocks
    end if
  end for
  $x \leftarrow$ Conv2D($x, ; out_c hannels = 1, ; kernel = 3 \times 3, ; stride = 1, ; padding = 1$) ▷ patch-level score map
  return x
end procedure

Algorithm 2 ESRGAN Generator (with detailed RRDB)

procedure DENSEBLOCK($x, channels, growth = 32, num_l ayers = 5$)
  $inputs \leftarrow x$
  for $i = 1$ **to** $num_l ayers$ **do**
    $conv_o ut \leftarrow$ Conv2D($inputs, ; out_c hannels = growth, ; kernel = 3 \times 3, ; padding = 1$)
    $act_o ut \leftarrow$ LeakyReLU($conv_o ut, ; slope = 0.2$)
    $inputs \leftarrow$ Concat($inputs, ; act_o ut$) ▷ channel-wise concat
  end for
  $proj \leftarrow$ Conv2D($inputs, ; out_c hannels = channels, ; kernel = 1 \times 1$) ▷ 1x1 projection
  return $x + 0.2 \times proj$ ▷ local residual scaling
end procedure
procedure RRDB($x, channels$)
  $out_1 \leftarrow$ DenseBlock($x, channels$)
  $out_2 \leftarrow$ DenseBlock($out_1, channels$)
  $out_3 \leftarrow$ DenseBlock($out_2, channels$)
  return $x + 0.2 \times out_3$ ▷ residual-in-residual with scaling
end procedure
procedure ESRGANGENERATOR($x, ; N_{RRDB} = 23, ; channels = 64$)
  $fea \leftarrow$ Conv2D($x, ; in = 3, ; out = channels, ; kernel = 3 \times 3, ; padding = 1$) ▷ shallow feature
  $trunk_i n \leftarrow fea$
  for $k = 1$ **to** N_{RRDB} **do**
    $trunk_i n \leftarrow$ RRDB($trunk_i n, ; channels$) ▷ stack of RRDB blocks
  end for
  $trunk_o ut \leftarrow$ Conv2D($trunk_i n, ; out = channels, ; kernel = 3 \times 3, ; padding = 1$)
  $fea \leftarrow fea + trunk_o ut$ ▷ global residual connection
  $fea \leftarrow$ Conv2D($fea, ; out = channels4, ; kernel = 3 \times 3, ; padding = 1$)
  $fea \leftarrow$ PixelShuffle($fea, ; upscale = 2$)
  $fea \leftarrow$ LeakyReLU($fea, ; slope = 0.2$)
  $fea \leftarrow$ Conv2D($fea, ; out = channels4, ; kernel = 3 \times 3, ; padding = 1$)
  $fea \leftarrow$ PixelShuffle($fea, ; upscale = 2$)
  $fea \leftarrow$ LeakyReLU($fea, ; slope = 0.2$) ▷ total $\times 4$ upscaling
  $out \leftarrow$ Conv2D($fea, ; out = 3, ; kernel = 3 \times 3, ; padding = 1$)
  $out \leftarrow$ Clamp$\big((out + 1)/2, ; 0, ; 1\big)$ ▷ map to [0,1] if original was [-1,1]
  return out
end procedure

Algorithm 3 VGG19 Feature Extractor (Perceptual Loss)

```
procedure VGGFeatureExtractor(x, ; target_layer =' relu5'_4)
    vgg ← Load pre-trained VGG19 features (FP)
    feature_extractor ← []
    for each layer index i in vgg sequential modules do
        append vgg[i] to feature_extractor
        if i corresponds to 'relu5_4' (e.g., index 35) then
            break
        end if
    end for
    Freeze parameters of feature_extractor (requires_grad = False)
    return feature_extractor(x)        ▷ feature maps for perceptual loss
end procedure
```

1) Algorithm:

Training Loop: The training was conducted in a mini-batch manner, using a batch size of 8 to balance computational efficiency and gradient stability. For each iteration, a batch of $64 \times 64 \times 3$ CLAHE-enhanced LR patches was fed into the generator, which produced the corresponding super-resolved images of size $256 \times 256 \times 3$.

These generated images were then passed to the discriminator, which attempted to distinguish between real HR patches and synthetic ones, enforcing the adversarial learning framework (1) (7). The generator was optimized using a weighted combination of pixel-wise L1 loss [23] (5), perceptual loss (computed from VGG19 feature space [24])(4), and adversarial loss (2) (6) to achieve both texture realism and structural fidelity. The discriminator was trained simultaneously using binary cross-entropy loss (3) [25] to correctly classify real and generated samples.

The model was trained using the Adam optimizer [26] (10) with a learning rate of 1×10^{-4} and $\beta_1 = 0.9$, $\beta_2 = 0.999$, for several epochs until convergence. This batch-wise adversarial training made the ESRGAN to effectively reconstruct fine retinal vessel textures and suppress over-smoothing artifacts common in interpolation-based methods.

Loss Functions:

$$\mathcal{L}_{adv}^{D} = \frac{1}{2}\left[\text{BCE}(D(I_{HR}), 1) + \text{BCE}(D(G(I_{LR})), 0)\right] \quad (1)$$

$$\mathcal{L}_{adv}^{G} = \text{BCE}(D(G(I_{LR})), 1) \quad (2)$$

where the Binary Cross-Entropy (BCE) loss is defined as:

$$\text{BCE}(p, y) = -\left[y \cdot \log(\sigma(p)) + (1-y) \cdot \log(1-\sigma(p))\right] \quad (3)$$

and $\sigma(p)$ denotes the sigmoid activation function.

$$\mathcal{L}_{perc} = \frac{1}{N}\sum_{i=1}^{N} \|\phi(I_{HR})_i - \phi(G(I_{LR}))_i\|_1 \quad (4)$$

where $\phi(\cdot)$ represents the VGG19 feature extractor (up to layer `relu5_4`), and $\|\cdot\|_1$ denotes the L_1 norm.

$$\mathcal{L}_{pixel} = \frac{1}{N}\sum_{i=1}^{N} \|I_{HR,i} - G(I_{LR})_i\|_1 \quad (5)$$

This measures the pixel-level L_1 distance between the high-resolution ground truth image I_{HR} and the generated image $G(I_{LR})$.

$$\mathcal{L}_G = \lambda_{adv}\,\mathcal{L}_{adv}^{G} + \lambda_{perc}\,\mathcal{L}_{perc} + \lambda_{L1}\,\mathcal{L}_{pixel} \quad (6)$$

where

$$\lambda_{adv} = 0.001, \quad \lambda_{perc} = 0.006, \quad \lambda_{L1} = 1.0$$

$$\mathcal{L}_D = \frac{1}{2}\left(\mathcal{L}_{adv}^{D,real} + \mathcal{L}_{adv}^{D,fake}\right) \quad (7)$$

$$\mathcal{L}_{adv}^{D,real} = \text{BCE}(D(I_{HR}), 1) \quad (8)$$

$$\mathcal{L}_{adv}^{D,fake} = \text{BCE}(D(G(I_{LR})), 0) \quad (9)$$

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \\ \hat{m}_t &= \frac{m_t}{1-\beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t}, \\ \theta_{t+1} &= \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \end{aligned} \quad (10)$$

D. Tiny Multi Scale Attention Network - Model 1

For vessel segmentation, the high-resolution OCTA patches a lightweight Tiny Multi-Scale Attention U-Net (TinyMSANet) shown in fig 8 designed to segment vascular structures with low computational cost. The encoder–decoder framework captures multi-level spatial features through successive convolution and pooling layers, while skip connections preserve detailed localization. A Multi-Scale Attention (MSA) module shown in fig 9 in the bottleneck adaptively models channel-wise correlations between query, key, and value projections, allowing the network to focus on vessel-relevant pixels while suppressing background noise. During decoding, transposed convolutions and feature concatenation progressively restore high-resolution continuity, ensuring precise reconstruction of vascular patterns.

The model is called TinyMSANet because it is a compact architecture, with a limited encoder depth of feature channels from 64
$rightarrow$256. From an input size of 256
$times$256
$times$3, the network carries out three steps of downsampling to achieve a good balance between detail and speed. This greatly reduces parameters and memory usage for speedier training and inference while remaining at good segmentation accuracy. Even though this model is light in construction, multi-scale feature extraction and attention in the bottleneck make possible the capture of vessel structures down to the fine details and clearly separate arteries and veins. The Mean

Algorithm 4 TinyMSANet for Vessel Segmentation

Require: Input image I, learning rate η
Ensure: Segmented vessel map $\hat{Y}$

procedure TRAINTINYMSANET(I)
Encoder:
$E_1 = \text{DoubleConv}(I, 16)$
$P_1 = \text{MaxPool}(E_1)$
$E_2 = \text{DoubleConv}(P_1, 32)$
$P_2 = \text{MaxPool}(E_2)$
$E_3 = \text{DoubleConv}(P_2, 64)$
$P_3 = \text{MaxPool}(E_3)$
x
$B = \text{ConvBNReLU}(P_3, 256)$
$B = \text{MSAttention}(B)$
x
Decoder:
$U_3 = \text{UpConv}(B, 64)$
$D_3 = \text{DoubleConv}(\text{Concat}(U_3, E_3), 64)$
$U_2 = \text{UpConv}(D_3, 32)$
$D_2 = \text{DoubleConv}(\text{Concat}(U_2, E_2), 32)$
$U_1 = \text{UpConv}(D_2, 16)$
$D_1 = \text{DoubleConv}(\text{Concat}(U_1, E_1), 16)$
x
Output Layer:
$\hat{Y} = \text{Conv}_{1\times1}(D_1, \text{out_channels})$
Loss: $\mathcal{L} = \text{MSELoss}(\hat{Y}, Y)$
return $\hat{Y}, \mathcal{L}$
end procedure

Squared Error MSE as loss function, is utilized as a training loss that prompts pixel-level fidelity to encourage smooth boundary reconstruction in the derived segmentation maps and hence ensure an anatomically coherent and accurate vessel delineation across the retinal vasculation.

1) Algorithm:

2) Training Loop: The TinyMSANet is trained for using the Adam optimizer with a learning rate of 1×10^{-4}. A batch size of 8 was used for mini-batch optimization. The Mean Squared Error (MSE)(11) loss was used as the training criterion. During each epoch, the network parameters were updated iteratively based on the computed loss for each batch, and all loss values were recorded to monitor training progression and stability. *TinyMSANet Loss*

$$\text{Mean Square Error } \mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{Y}_i - Y_i\right)^2 \tag{11}$$

E. Tiny Multi Scale Attention Network - Model 2

The second TinyMSANet model is developed for vessel refinement and selective enhancement, for post-processing stage to improve the quality initial segmentation. It takes as input the segmentation output from the first TinyMSANet alongside the corresponding ESRGAN-enhanced OCTA image, where the green-channel regions representing capillaries were suppressed.This shifts the attention of the network to the more salient arteries and veins while reducing the effect of the finer capillary structures that introduce noise or oversegmentation. The model re-examines spatial hierarchies and strengthens vessel continuity by selectively emphasizing relevant vascular signals through the encoder-decoder pathway. Further, it refines discrimination between arteries and veins by accentuating stronger vascular responses with suppression of residual background variations. Training is performed using Mean Squared Error (MSE) loss against capillary-suppressed ground truth maps, to make sure of anatomically consistent refinement that preserves the integrity of major vessels while effectively reducing microvessel over-segmentation.

IV. EXPERIMENTS AND RESULTS

This section describes the experiments conducted to evaluate the performance of the proposed framework along with the underlying frameworks.

A. Experimental Setup

The experiments are conducted using the publicly available OCTA-500 dataset [27]. Preprocessing involves applying Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 8, followed by cropping each image to a size of $256 \times 256 \times 3$ and subsequently resizing it to $64 \times 64 \times 3$ and persist the capability of dynamic augmentation. The image enters ESRGAN model for image reconstruction and enhancement. Training has been performed on the GPU environment utilizing a VGG19-based feature extractor. The model is optimized using a standard configuration of loss functions and optimizers.Evaluation is done with several quantitative metrics on pixel level accuracy and perceptual quality. The generated image processeds for TinyMSANet architecture trained for fine vessel segmentation and refinement. After the image get segmented by the first TinyMSANet. Then the green channel suppresses to enhance the visibility and separation of arteries and veins structure passes to second TinyMSANet model. The configuration of training in this stage remains the same compared to the previous stages; optimizers and performance evaluation metrics remain identical across models. A summary of all key parameter configurations used across the three stages are provided in Table II.

B. Performance Metrics

The performance of the ESRGAN and the proposed TinyMSA-Net models (TinyMSA-Net 1 and TinyMSA-Net 2) is briefly defined below along with its standard mathematical formulation.

1) Mean Absolute Error (MAE): MAE is the average absolute difference between predicted and ground truth values, indicating the overall prediction accuracy.

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{12}$$

TABLE II: **Parameter setup for the three-stage experimental pipeline**

Parameter	**ESRGAN**	**TinyMSANet-1**	**TinyMSANet-2**
Input size	$64 \times 64 \times 3$	$256 \times 256 \times 3$ (ESRGAN output)	$256 \times 256 \times 3$ (TinyMSANet-1 output)
Feature extractor	VGG19-based	–	–
Loss functions	L_1 Loss, BCEWithLogitsLoss	Mean Squared Error (MSE) Loss	Mean Squared Error (MSE) Loss
Optimizer	Adam	Adam	Adam
Learning rate	1×10^{-4}	1×10^{-4}	1×10^{-4}
β **values**	$(0.9, 0.999)$	$(0.9, 0.999)$	$(0.9, 0.999)$
Batch size	8	8	8
Epochs	47	30	19
Number of images	2100 (Dynamic Augmentation)	2100	2100

2) Mean Squared Error (MSE): MSE is the mean of squared differences between the predicted and actual pixel intensities, penalizing larger errors more heavily.

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{13}$$

3) Dice Coefficient / Dice Loss: The Dice Coefficient [28] is the spatial overlap between predicted and ground truth segmentation masks. Dice Loss is derived as $1 - \text{Dice}$.

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|} \tag{14}$$

4) Jaccard Index / Jaccard Loss: The Jaccard Index [29], also known as the Intersection-over-Union (IoU), it is the ratio of overlap to the union of predicted and ground truth regions. Jaccard Loss is computed as $1 - \text{Jaccard}$.

$$\text{Jaccard} = \frac{|A \cap B|}{|A \cup B|} \tag{15}$$

5) Structural Similarity Index (SSIM): SSIM tells about the image quality by comparing luminance, contrast, and structural information between two images.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{16}$$

6) Learned Perceptual Image Patch Similarity (LPIPS): LPIPS is the perceptual similarity by comparing deep feature representations of two images extracted from pretrained networks.

$$\text{LPIPS}(x, y) = \sum_{l} \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (\hat{y}_{hw}^l - y_{hw}^l)\|_2^2 \tag{17}$$

7) Peak Signal-to-Noise Ratio (PSNR): PSNR [30] is a ratio between the maximum possible pixel intensity and the mean squared error, expressed in decibels (dB).

$$\text{PSNR} = 10 \log_{10}\left(\frac{MAX_I^2}{\text{MSE}}\right) \tag{18}$$

These parameters collectively provide a comprehensive evaluation of model accuracy, perceptual quality, and structural fidelity across the three network variants.

C. Result Analysis

The ESRGAN model outperforms a standalone SRGAN [21], ESRGAN [22] and its sharpened versions, as presented in Table III. Such enhanced ESRGAN output produces sharper images, smoother vessel edges, and increased visibility of the retinal structures which is very important in medical image processing. Such improvements can help to conduct more accurate vessel detection. Furthermore, the enhanced images are better suited for the next stage of segmentation. Training loss is presented in Fig. 11.

The proposed TinyMSANet model, summarized in Table V, uses these ESRGAN-enhanced images to produce more accurate and stable segmentation. Among all its versions, the stacked TinyMSANet gives the best results across MAE, Dice, and PSNR when compared to U-Net [31] and Attention U-Net [32]. This shows its strength in capturing multi-scale vessel features and improving artery–vein segmentation. Together, the ESRGAN enhancement and TinyMSANet segmentation form a strong and reliable pipeline for retinal image analysis. The confusion matrix [33] and ROC curves [34] are shown in Fig. 12 and Fig. 13.

V. Scalability

To check how well the TinyMSANet framework scales and generalizes, two types of experiments were done. First, the trained models were tested on different datasets to measure their ability to generalize. Second, tests were run using different dataset sizes to see how the model's performance changes as the amount of training data increases.

A. Cross-Dataset Evaluation

The models were trained and tested on the OCTA-500 dataset (Fig. 14) which are further tested on the ROSE dataset

TABLE III: Quantitative Comparison between SRGAN and ESRGAN Models

Model	MAE	MSE	Dice Loss	Jaccard Loss	SSIM	LPIPS	PSNR
SRGAN (Normal)	0.1765	0.0504	0.5816	0.7354	0.2117	0.4817	12.98
SRGAN (Sharpen)	0.3359	0.1644	0.5818	0.73	0.0955	0.7680	7.84
ESRGAN (Normal)	0.1018	0.0175	0.5794	0.73	0.3939	0.457	17.59
ESRGAN (Sharpen)	0.1247	0.0278	0.5755	0.73	0.3872	0.4734	15.56

TABLE IV: Quantitative Comparison between TinyMSA-Net Models

Model	MAE	MSE	Jaccard Index	Dice Coeff.	PSNR
TinyMSANet	0.1334	0.0647	0.3188	0.4823	11.99
TinyMSANet Capillary Removed	0.0251	0.0095	0.6251	0.7681	20.27
TinyMSANet Stacked	0.0201	0.0091	0.6546	0.7906	20.38

TABLE V: Quantitative Comparison of Different Models on Retinal Vessel Segmentation

Model	MAE	MSE	Jaccard Index	Dice Coeff.	PSNR
UNet	0.1384	0.0662	0.2920	0.4520	11.89
Stacked UNet	0.0249	0.0109	0.5793	0.7309	19.71
Attn UNet	0.1611	0.0667	0.2925	0.4517	11.85
Stacked Attn UNet	0.0580	0.0134	0.5921	0.7415	18.78
TinyMSANet	0.1334	0.0647	0.3188	0.4823	11.99

without any extra fine-tuning. This cross-dataset test verifies the performance of the network against major changes in image resolution, lighting, and vessel patterns. The similar visual and numerical performance on the ROSE dataset [35] [36] demonstrates the robustness, adaptability, and clinical usability of TinyMSANet under different imaging conditions.

B. Evaluation on Varying Dataset Sizes

To further test how well TinyMSANet scales with larger and more varied data, three subsets of the OCTA-500 dataset—Small, Medium, and Large—were used in the experiments. The results in Table VI show steady performance gains as the dataset size increases, proving that the model scales effectively and benefits from more training data.

VI. Adaptability Analysis

The proposed TinyMSANet framework shows good adaptability for different segmentation tasks. Its modular design allows easy changes to the loss function and activation functions based on the goal and dataset. This section discusses three possible extensions of the framework: stacked TinyMSANet, instance-level segmentation [37], and modifications to the activation function.

A. Ablation Study

Compared the proposed TinyMSANet with the Stacked TinyMSANet to investigate how different variants of an architecture affect its performance. It is evident that the stacked version considerably enhanced feature aggregation through multi-level refinement across successive stages, therefore reconstructing fine-grained vascular patterns in a superior way. Looking at Table VII, it is apparent that the Stacked TinyMSANet continually outperforms TinyMSANet across all metrics while achieving substantial gains on Jaccard Index (+30.6%), Dice Coefficient (+28.6%), and PSNR (+8.28 dB), thus proving that stacking MSANet modules allows deeper contextual reasoning and more stable convergence during training.

To extend the framework to instance-level vessel segmentation, TinyMSANet is modified by replacing the pixel-level MSE loss with Cross-Entropy Loss and by generating masks that include clear instance IDs. This change helps the model separate vessels that overlap or are very close to each other while keeping the vessel structure consistent. Table VIII shows the difference between the original TinyMSANet and the Instance TinyMSANet version. Although instance-aware learning improves category-level separation (seen through higher Jaccard and Dice scores), it causes a small increase in pixel-level error because instance boundaries are more complex.

B. Hyperparameter Optimization

The baseline TinyMSANet architecture utilizes the ReLU [38] activation function for its computational efficiency and fast convergence. However, substituting ReLU or LeakyReLU [39] with a zero-centered activation function such as *tanh* can enhance gradient flow across layers and potentially improve the network's representational capability. While this modification may lead to improved feature smoothness and

TABLE VI: Performance Comparison of TinyMSANet and Stacked TinyMSANet on Different Dataset Sizes

Model	Dataset Size	MAE	MSE	Jaccard Index	Dice Coefficient	PSNR
TinyMSANet	Small	0.1424	0.0623	0.3298	0.4853	11.06
TinyMSANet	Medium	0.1334	0.0647	0.3188	0.4823	11.99
TinyMSANet	Large	0.1334	0.0624	0.3198	0.4035	11.25
Stacked TinyMSANet	Small	0.0250	0.0095	0.6260	0.7678	20.26
Stacked TinyMSANet	Medium	0.0201	0.0095	0.6251	0.7681	20.27
Stacked TinyMSANet	Large	0.0211	0.0095	0.6243	0.7662	20.21

TABLE VII: Performance Comparison: TinyMSANet vs. Stacked TinyMSANet

Model	MAE	MSE	Jaccard	Dice	PSNR
TinyMSANet	0.1334	0.0647	0.3188	0.4823	11.99
Stacked TinyMSANet	0.0201	0.0095	0.6251	0.7681	20.27

TABLE VIII: Performance Comparison: TinyMSANet vs. Instance TinyMSANet

Model	MAE	MSE	Jaccard	Dice	PSNR
TinyMSANet	0.1334	0.0647	0.3188	0.4823	11.99
Instance TinyMSANet	0.0547	0.0878	0.5542	0.7131	10.56

TABLE IX: Performance Comparison: TinyMSANet vs. Instance TinyMSANet

Model	MAE	MSE	Jaccard	Dice	PSNR
TinyMSANet - ReLU	0.1334	0.0647	0.3188	0.4823	11.99
TinyMSANet - tanh	0.1247	0.0778	0.2942	0.5131	10.42

robustness, it introduces a slower convergence rate, thereby increasing the overall training time required to reach the global minimum. Shown in Table IX.

VII. Comparative Study

Most existing work on retinal vessel segmentation focuses only on binary segmentation [31], as shown in Table X. In contrast, the proposed model performs multi-class segmentation, with clear separations for arteries and veins. This creates a very specific and clinically useful understanding of the retinal structures.

The framework also comprises a generator network that enhances the resolution of significant regions. This improves the quality of the input images before segmentation, helping the model achieve more precise and efficient results. The model is lightweight, having fewer encoder and decoder layers and fewer parameters. Even so, it has achieved accuracy and loss values comparable to deeper and more complex binary models. This means that the method will be both effective and efficient, providing quality segmentation with only a small and low-cost model. Overall, the proposed framework performs well and scales reliably; hence, it is suitable for real-time and resource-limited medical applications.

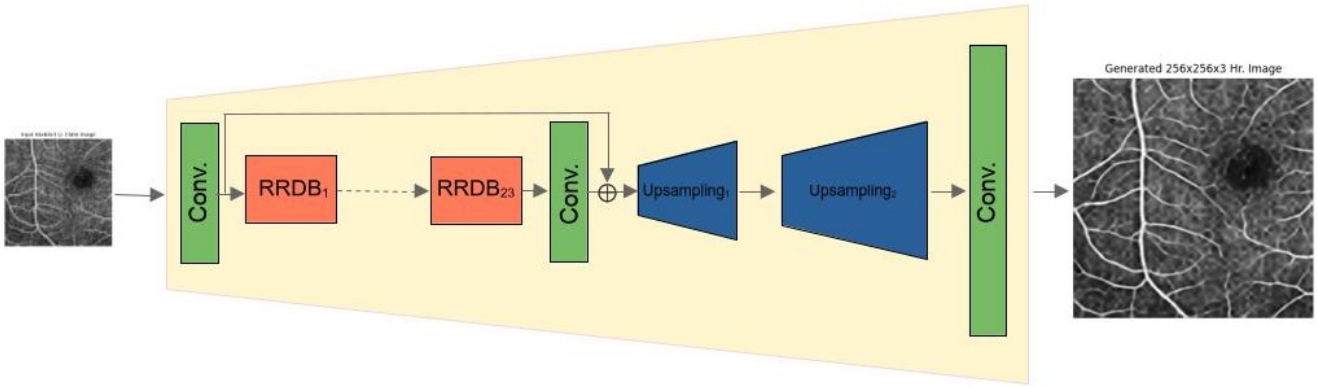
VIII. Conclusion

The whole idea behind this framework is to keep the focus on necessary regions of a medical image rather than process the complete image all at once. By doing so, it makes the generator module flexible-it enables enhancing the resolution of any selected region, such as a $64 \times 64 \times 3$ patch, where more visual detail is needed. The same architecture for segmentation is able to nicely separate the arteries and veins. The model can even work directly on full $256 \times 256 \times 3$ OCTA images, thereby showing that this model is adaptable with different resolutions. The architecture runs efficiently with low computation and lesser parameters, which makes it lightweight and practical for real-world medical imaging segmentation.

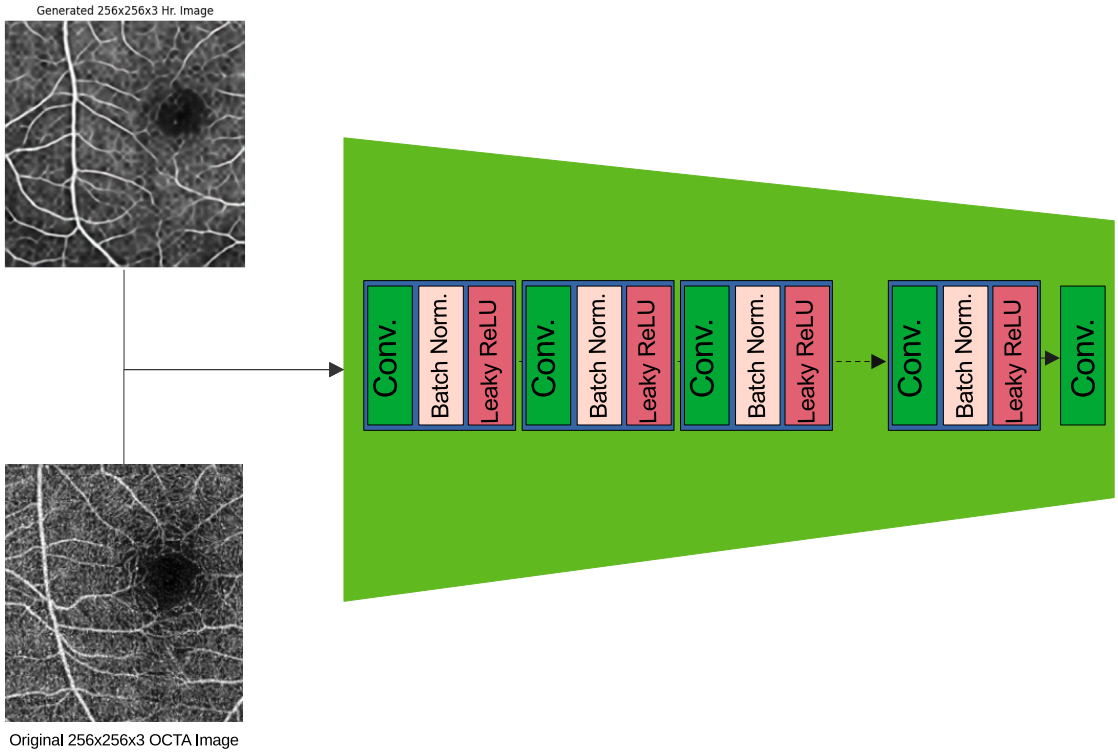
In the future, the framework can be extended to full 3D OCTA volumes.Domain adaptation can make it more robust across different scanners and patient groups. Improved segmentation modules may also target finer structures such as capillaries, non-perfusion regions in order to increase clinical accuracy and real-world usefulness.

TABLE X: Comparison of Existing Methods with Proposed Approach

Author	**Method**	**Dataset**	**Dice Coefficient**	**PSNR / Accuracy**
Hao et al.	Sparse-based domain adaptation with GAN and edge loss	SURE-O, SURE-Z	0.601	74.7 (Accuracy)
Manassakorn et al.	Custom CNN (GlauNet) for glaucoma detection in OCTA	–	–	0.89 (Accuracy)
Li et al.	3D-to-2D deep learning with projection learning	–	0.88	–
Zhang et al.	OCT + OCTA fusion for fast artifact removal	–	–	16 (PSNR)
Hao et al.	VAFF-Net with multi-input en face slabs	–	0.81	–
Abdelsalam et al.	Multifractal geometry + SVM	–	–	98.5 (Accuracy)
Proposed	**ESRGAN sequenced to Dual TinyMSANet**	OCTA-500	**0.768**	**20.27 (PSNR)**

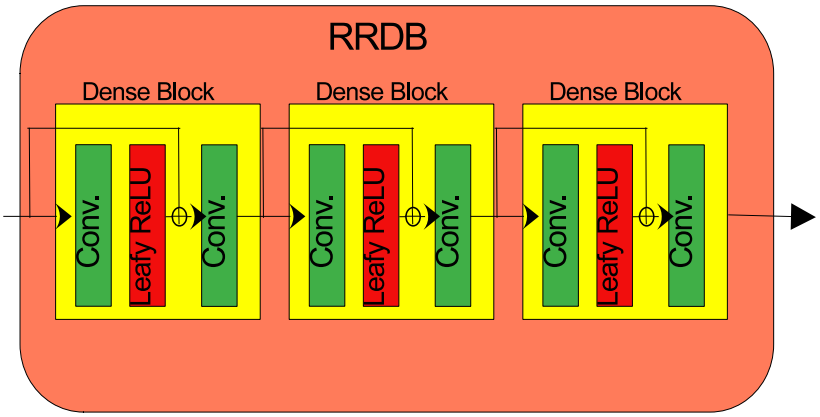


(a) Generator architecture.



(b) Discriminator architecture.

Fig. 5: Overview of ESRGAN components: (a) Generator, (b) Discriminator, (c) RRD Block architectures



(a) RRD Block.

(b) Overall dataflow of the proposed framework.

Fig. 6: Overview of components: (a) RRD Block, (b) Dataflow architectures

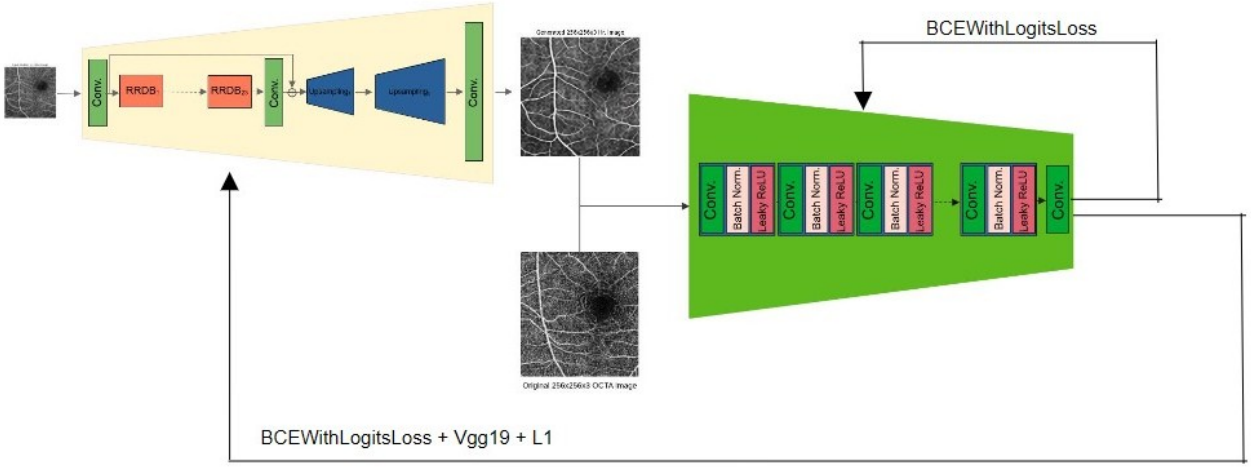


Fig. 7: Training of ESRGAN.

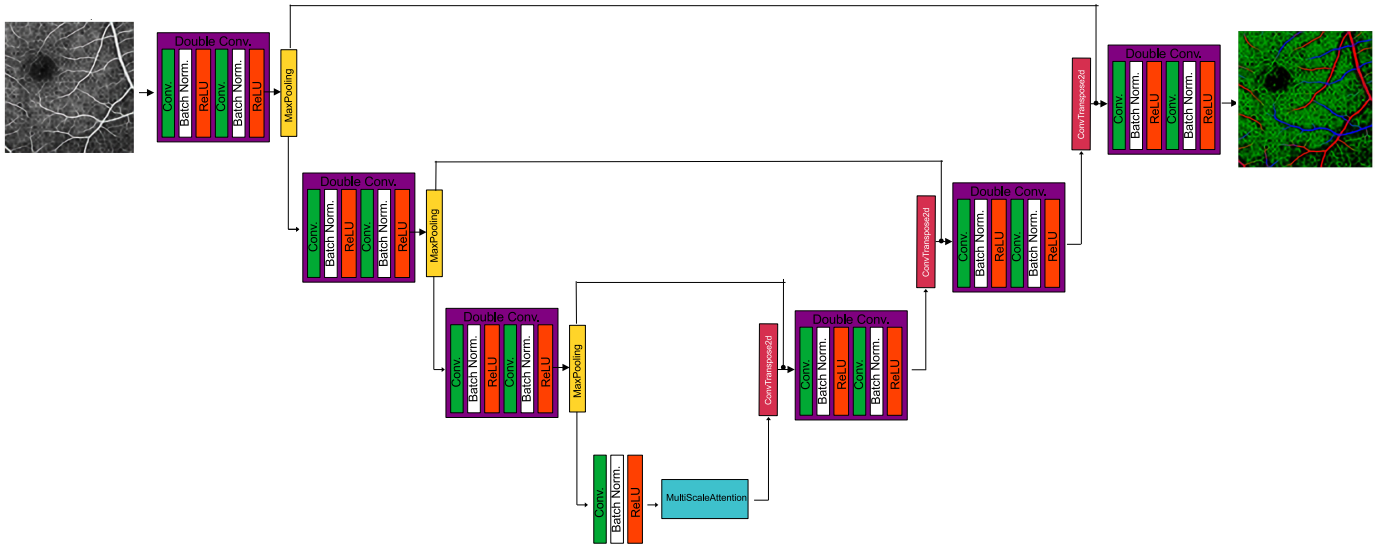


Fig. 8: Tiny MSANet

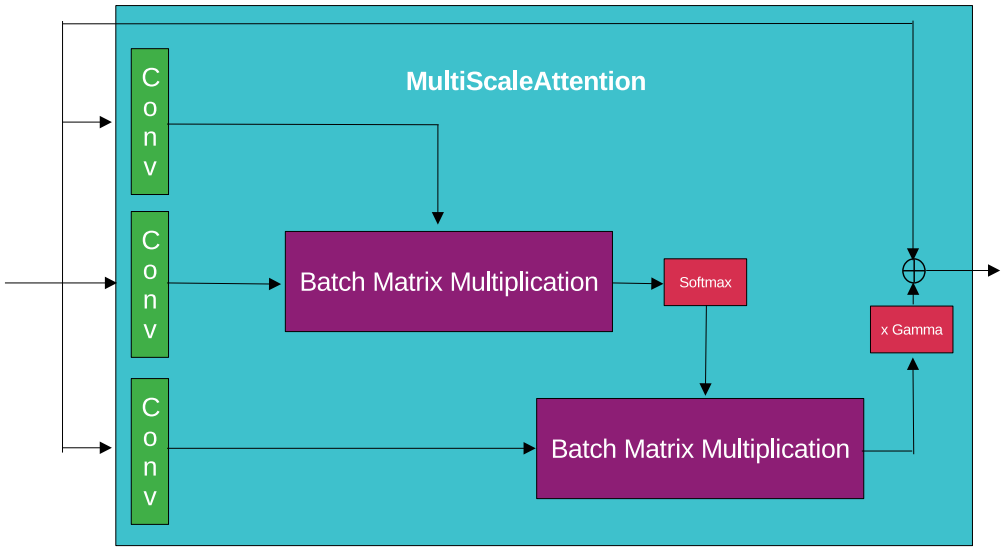
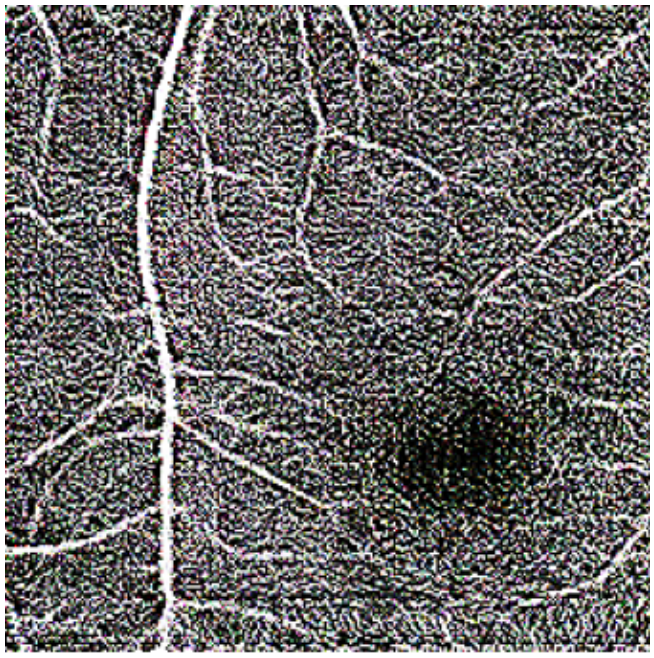
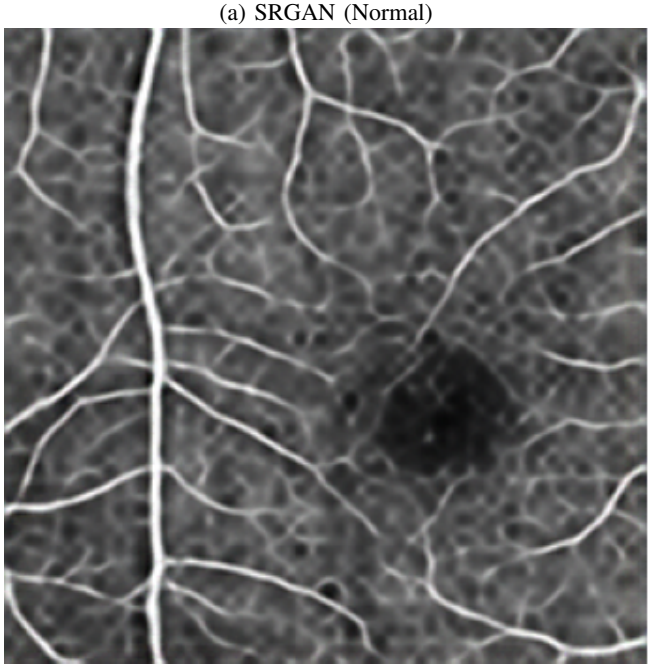


Fig. 9: Multi Scale Attention

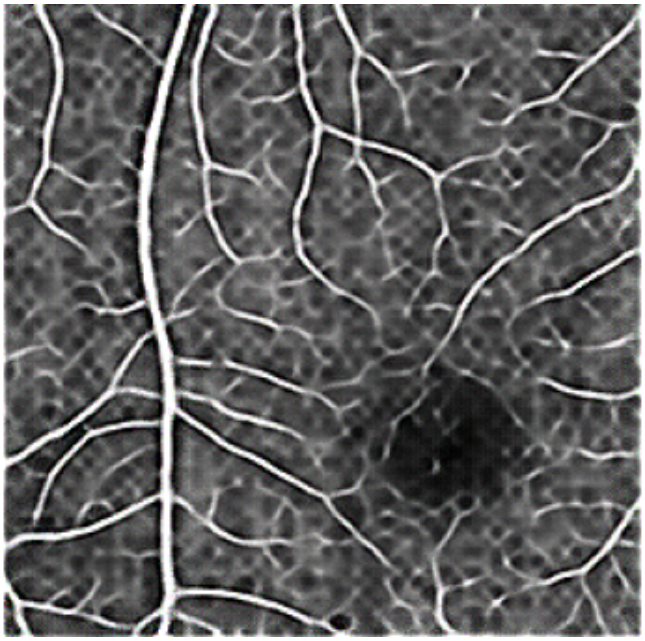
(a) SRGAN (Normal)



(b) SRGAN (Sharpen)



(c) ESRGAN (Normal)



(d) ESRGAN (Sharpen)

Fig. 10: Visual comparison of reconstructed outputs from SRGAN and ESRGAN models. Each subfigure corresponds to different sharpening configurations.

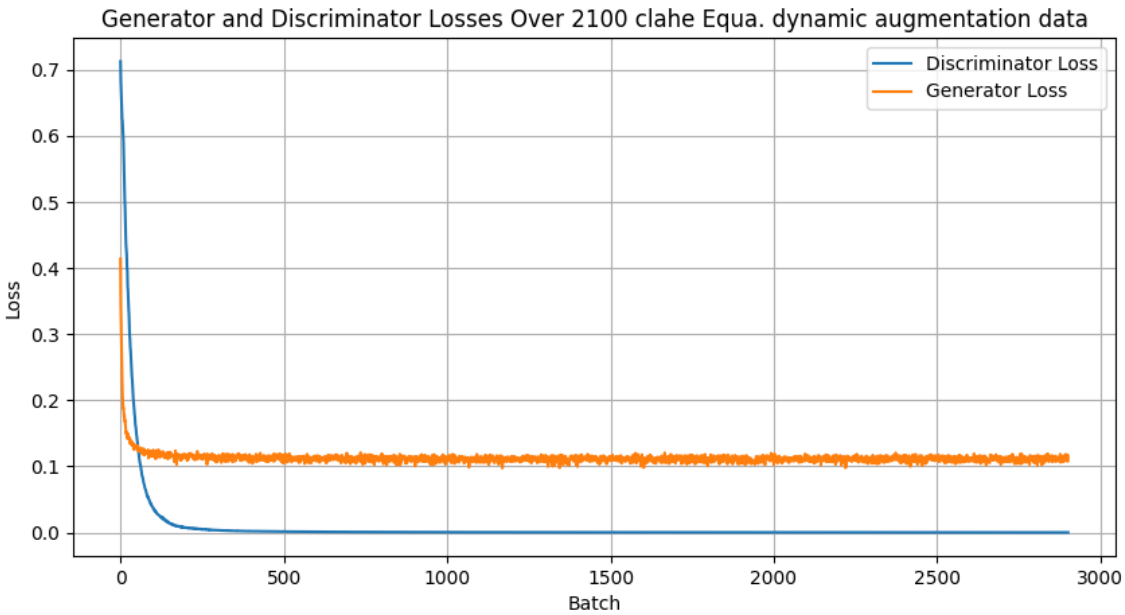


Fig. 11: Training Loss

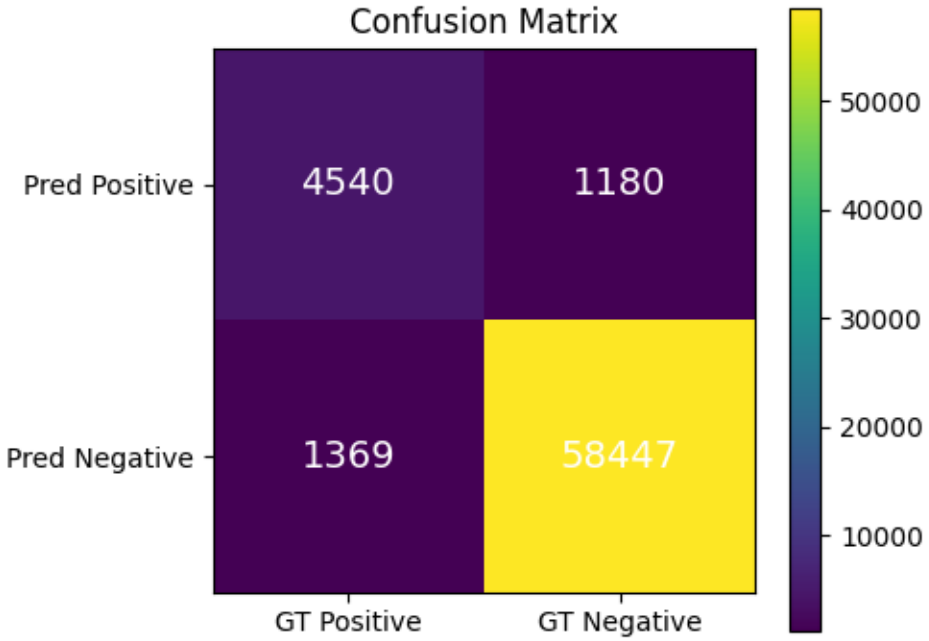


Fig. 12: Confusion matrix

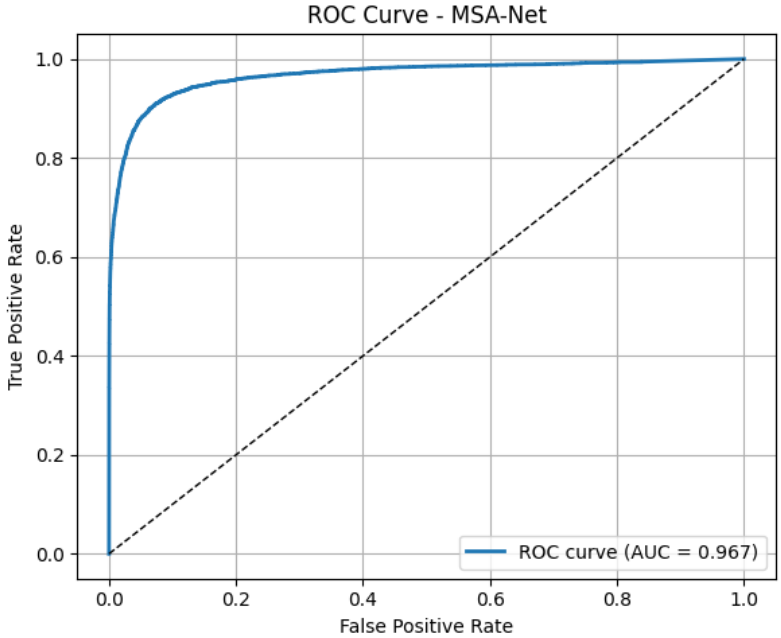
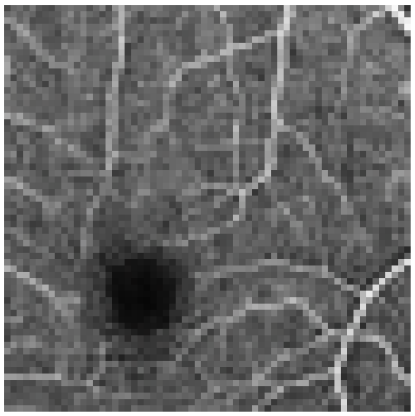
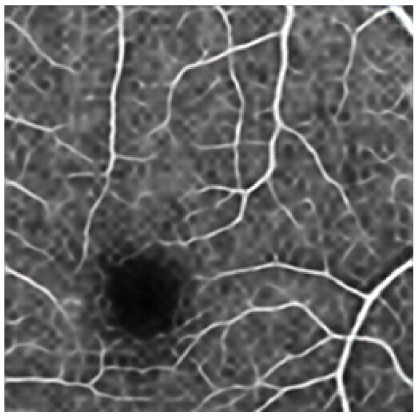


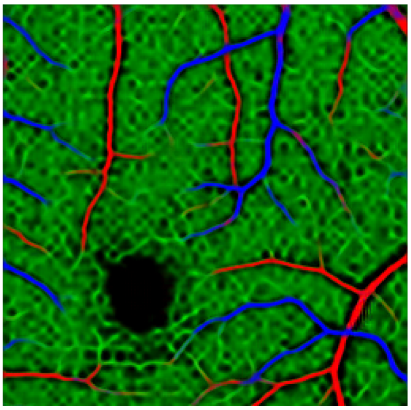
Fig. 13: ROC



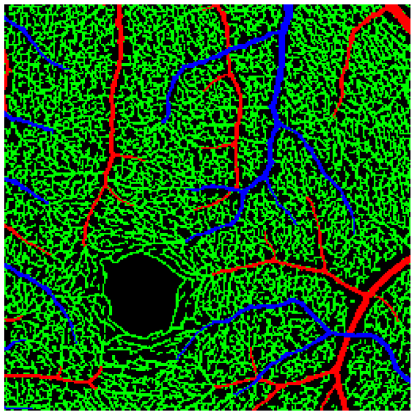
(a) Input Clahe



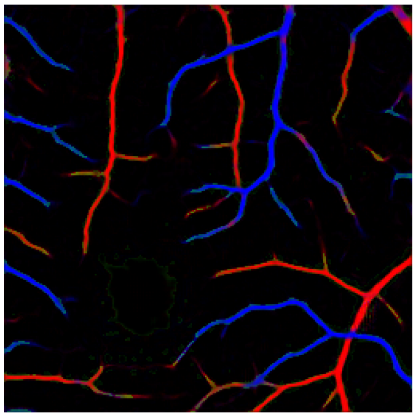
(b) Generated Image



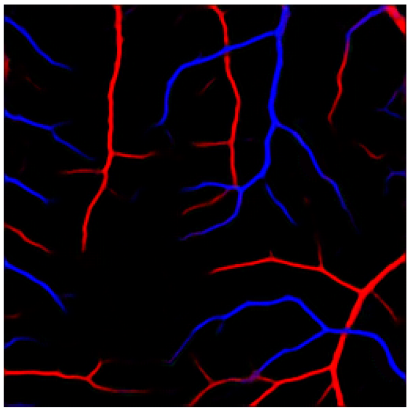
(c) TinyMSANet 1 Output



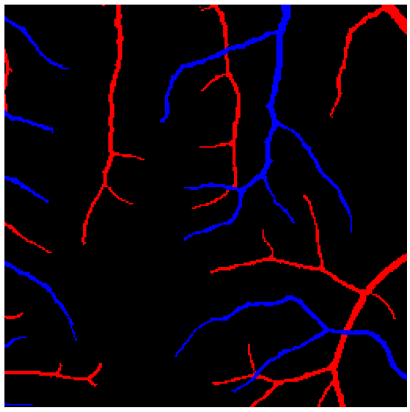
(d) Ground Truth



(e) TinyMSANet 1 Output Capillary

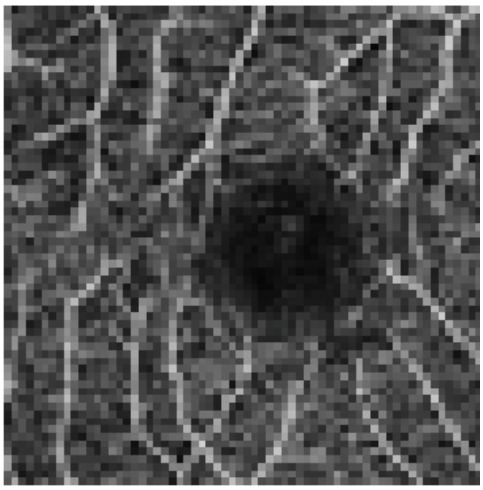


(f) TinyMSANet 2 Output

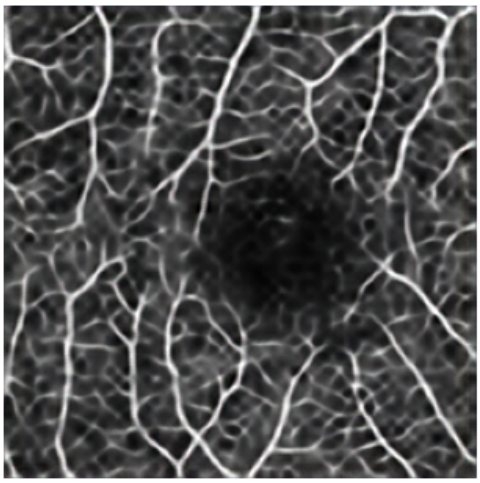


(g) Ground Truth Capillary Removed

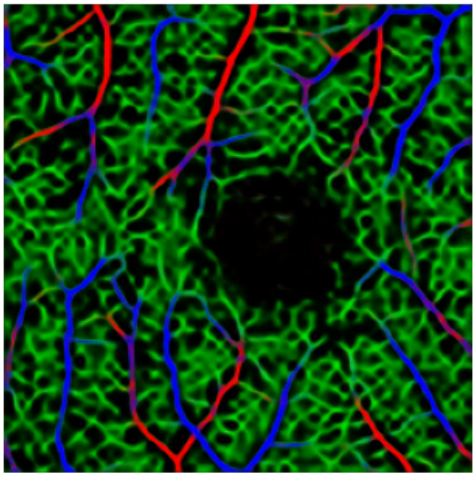
Fig. 14: Visual output comparison of the proposed TinyMSA-Net model variants. Each subfigure represents different configurations and ground-truth references used to evaluate segmentation scalability across datasets.



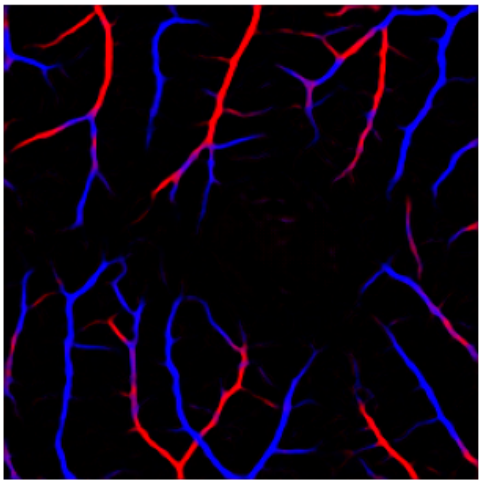
(a) Input CLAHE



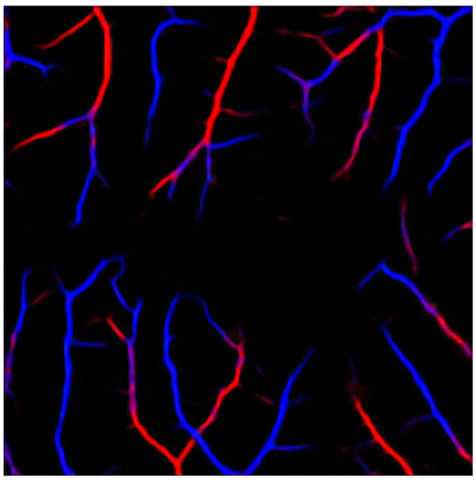
(b) Generated Image



(c) TinyMSANet 1 Output



(d) TinyMSANet 1 Output (Capillary Removed)



(e) TinyMSANet 2 Output

Fig. 15: Visual comparison of the proposed TinyMSA-Net model variants on the ROSE dataset, showing cross-dataset generalization performance.

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